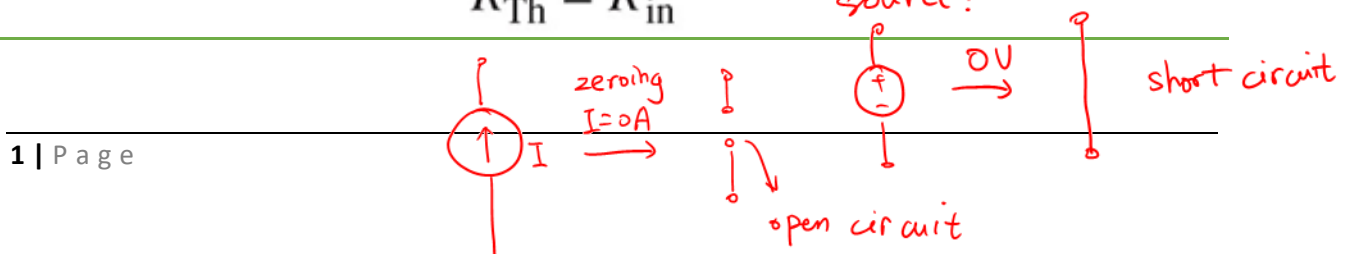
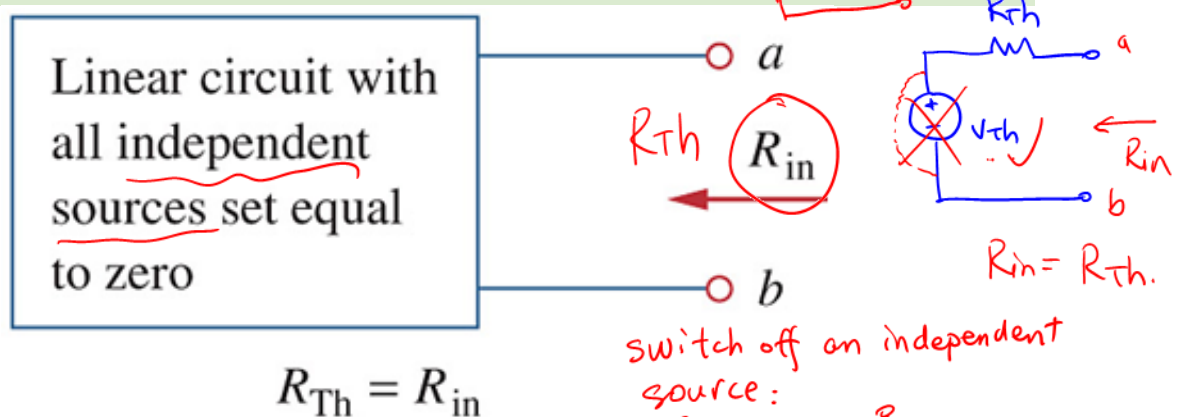
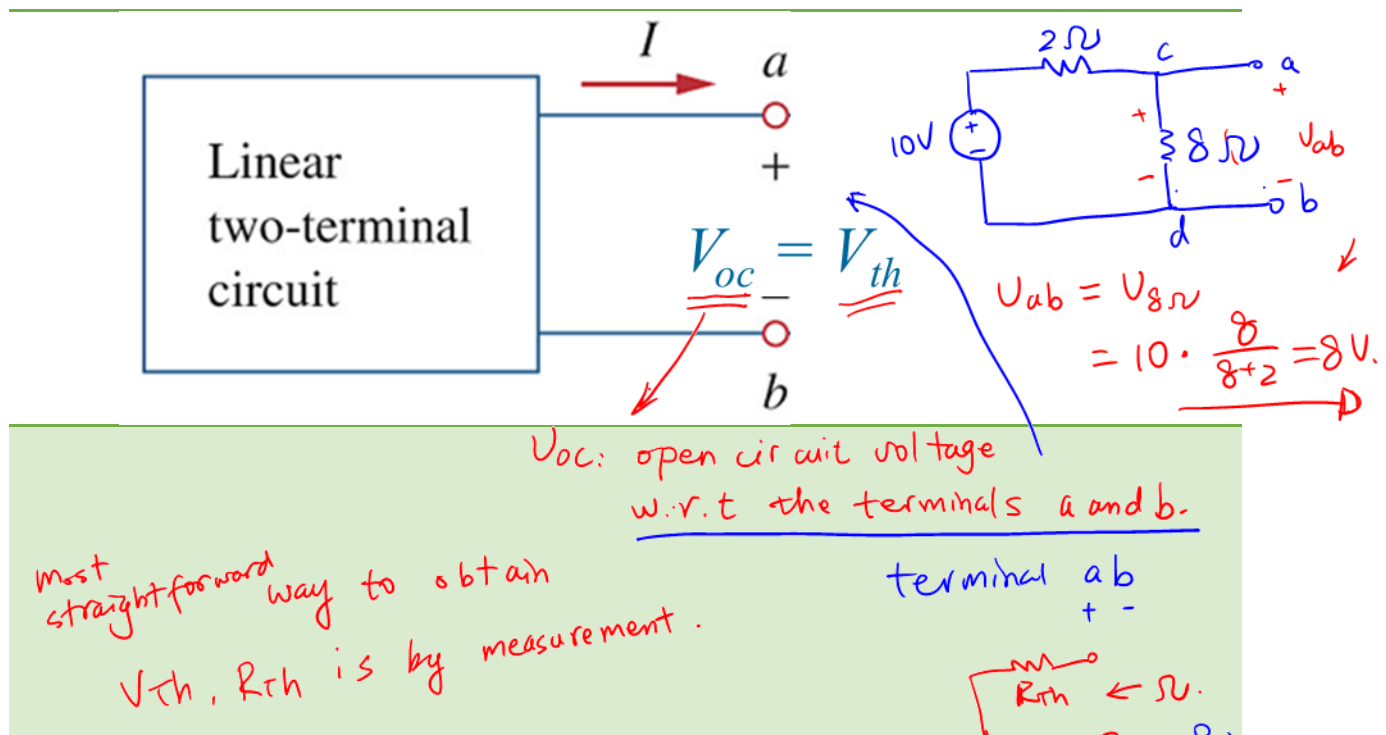
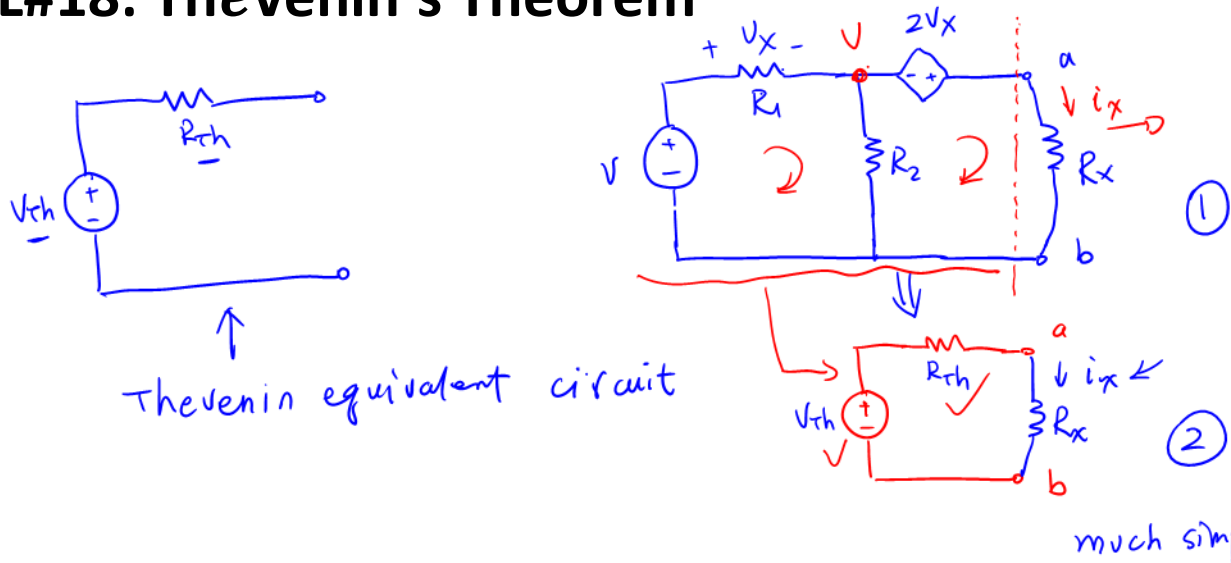
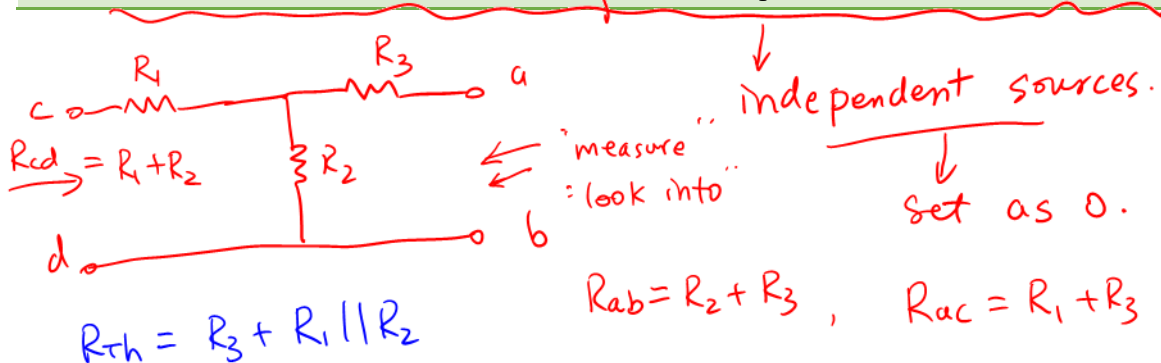


L#18: Thévenin's Theorem



Method #1: Zeroing of Sources:

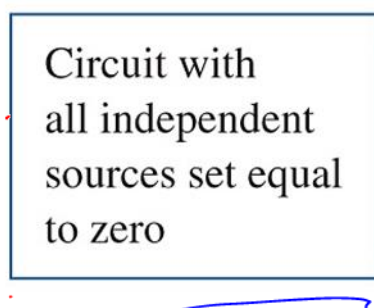
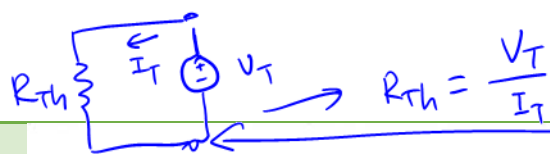
If the circuit contains no dependent sources.



Method #2: Application of test sources, if the circuit does contain dependent sources.

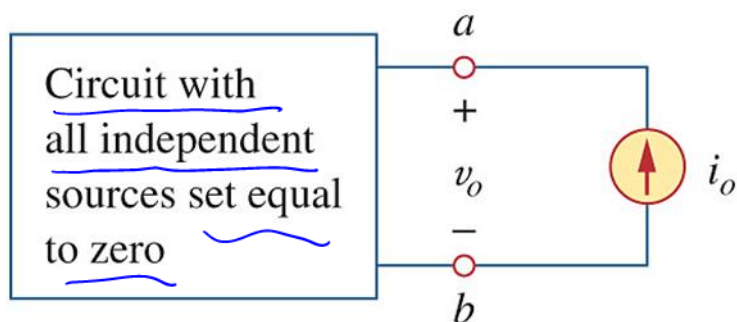
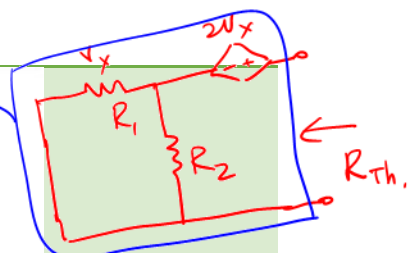
↓↓
we can't simply switch off the independent sources.

- ① switch off all ind. sources.
- ② all dependent sources remain unchanged.



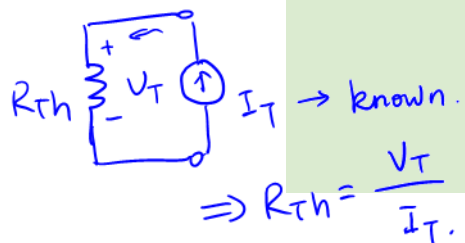
$$R_{Th} = \frac{v_o}{i_o}$$

(a)



$$R_{Th} = \frac{v_o}{i_o}$$

(b)

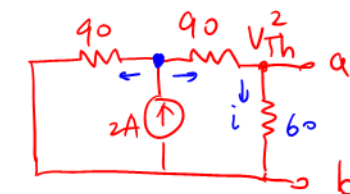


① ②

Example 8: Thévenin

- Obtain the Thévenin equivalent circuit to the left of terminals $a-b$.
- Determine current I .

contribution from
2A to V_{Th} , V_{Th}^2

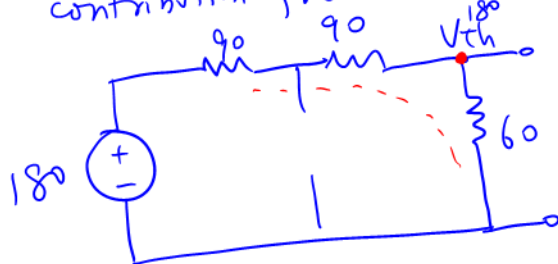


$$V_{Th}^2 = 60i = 60 \cdot 2 = 120$$

$$V_{Th}^2 = 120 \cdot \frac{90}{90+90+60} = 45V$$

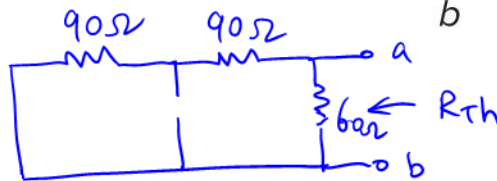
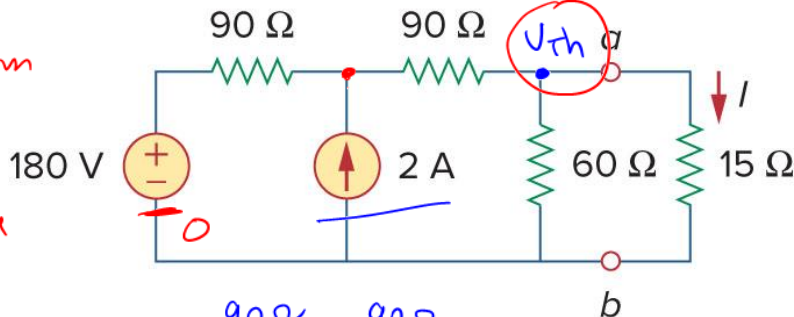
superposition:

contribution from 180V to V_{Th} , V_{Th}^{180}



$$\Rightarrow V_{Th}^{180} = 180 \cdot \frac{60}{60+90+90} = 45V$$

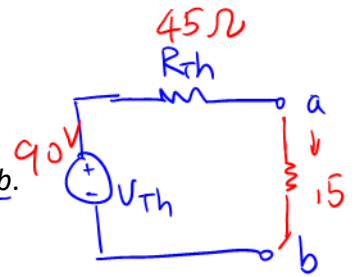
$$V_{Th} = V_{Th}^{180V} + V_{Th}^{2A} = 90V$$



$$\Rightarrow R_{Th} = 60 \parallel (90+90)$$

$$= \frac{60 \times 180}{60+180}$$

$$= \frac{180}{4} = 45\Omega$$

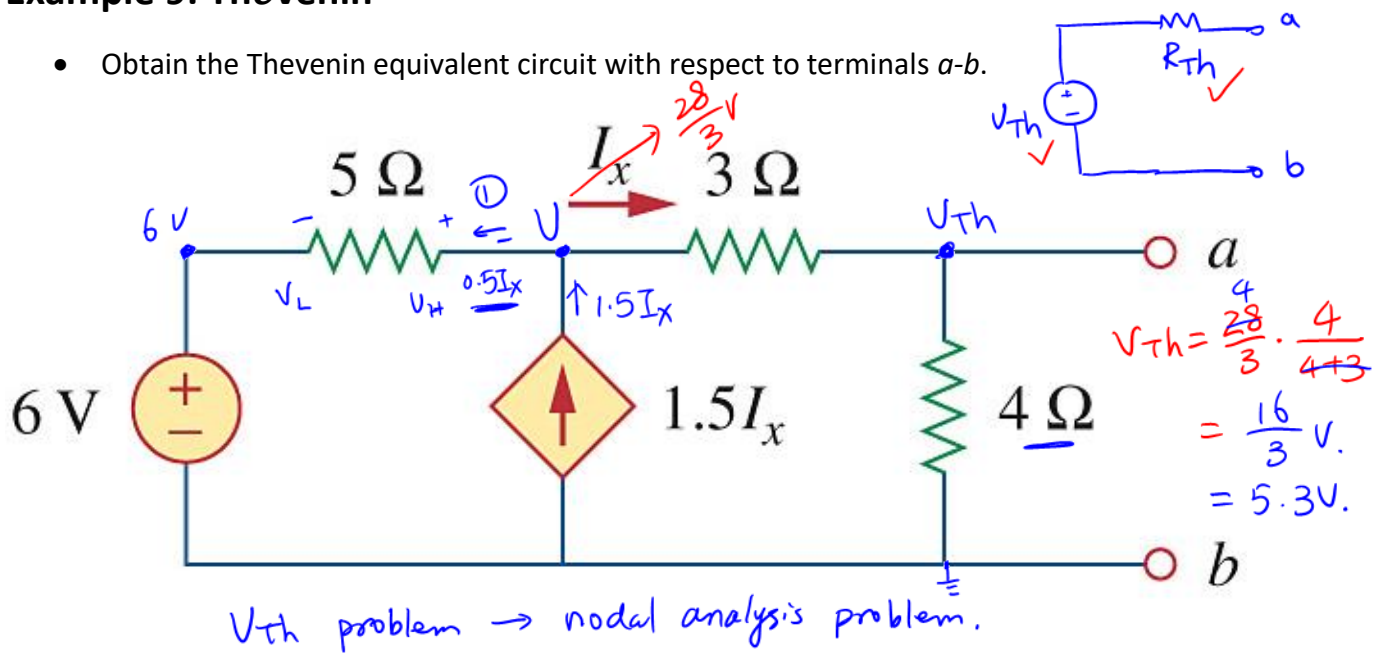


$$I = \frac{90}{45+15}$$

$$= 1.5A$$

Example 9: Thévenin

- Obtain the Thevenin equivalent circuit with respect to terminals $a-b$.



$$0.5 I_{\underline{x}} = \frac{\underline{v} - 6}{5} \quad (1)$$

$$I_x = \frac{V}{4+3} = \frac{V}{7} \quad (2)$$

② $\Rightarrow$ ①

$$\frac{v}{2 \times 7} = \frac{v-6}{5}$$

$$5V = 14V - 8.4$$

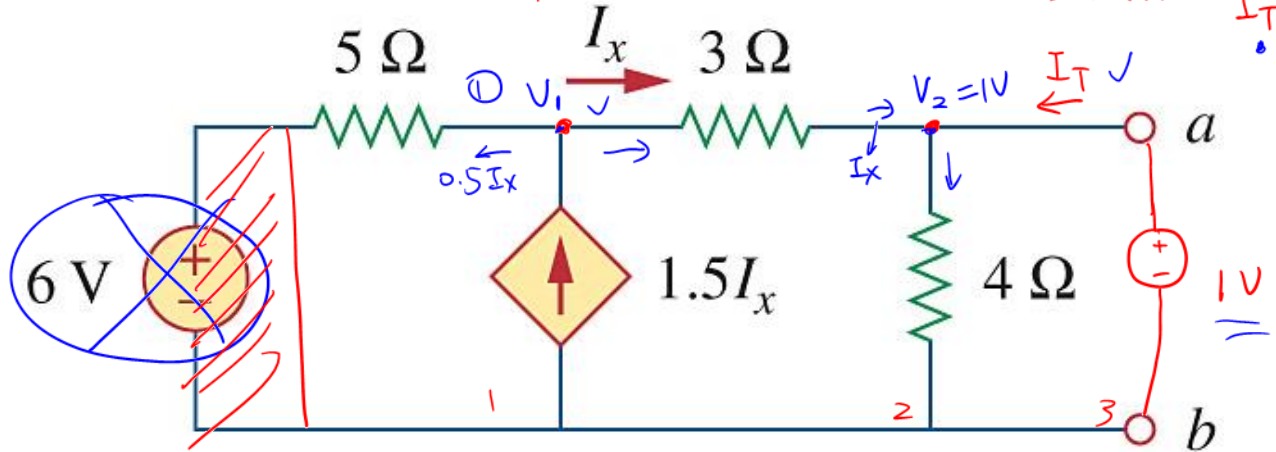
$$\Rightarrow v = \frac{84}{9} = \frac{28}{3} \text{ V.}$$

R_{Th}: ① switch off

② add test source. $\left\{ \begin{array}{l} V \\ A \end{array} \right.$

$$R_{Th} = \frac{1}{9/4} = \frac{4}{9} \Omega.$$

$$\leftarrow \Rightarrow R_{Th} = \frac{U_T = 1V}{I_T}$$



Add voltage source

$$0.5I_x = \frac{V_1}{5} \quad (1)$$

$$I_x = \frac{V_1 - V_2}{3} \quad (3)$$

$$I_x + I_T = \frac{V_2}{4} = \frac{1}{4} \quad (2)$$

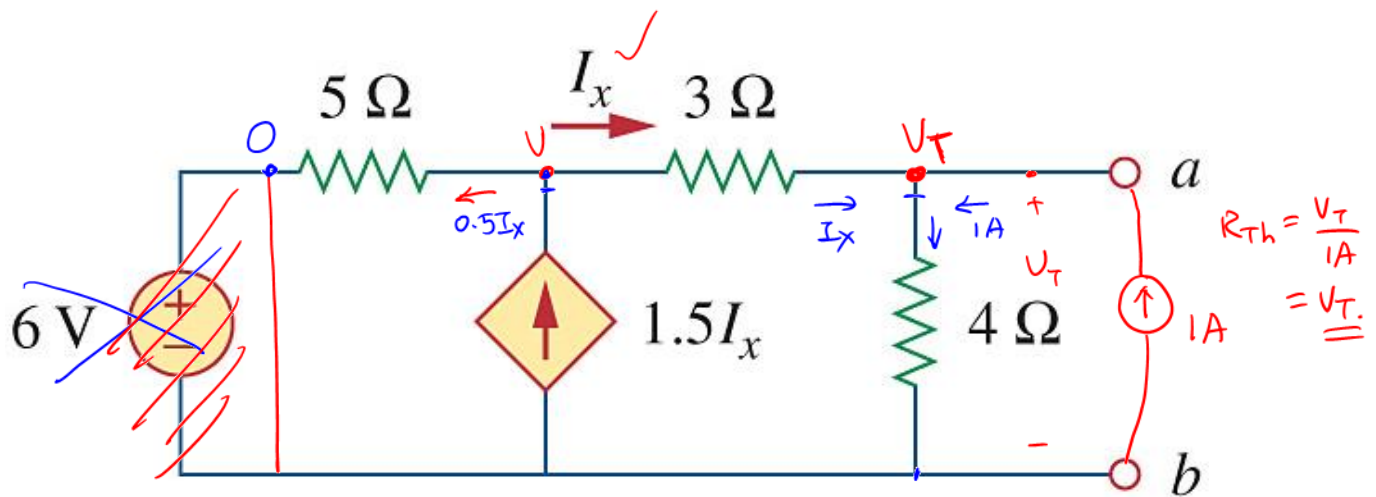
I_x, V_1, I_T D.

$$V_1 = 2.5I_x \quad (1)$$

$$3I_x = V_1 - 1 = 2.5I_x - 1$$

$$\Rightarrow 0.5I_x = -1, \Rightarrow \underline{I_x = -2A.}$$

$$\Rightarrow I_T = \frac{1}{4} - I_x = \frac{1}{4} - (-2) = \frac{9}{4} A.$$



Add current source

$$0.5I_x = \frac{V}{5} \Rightarrow V = 2.5I_x. \quad (1)$$

$$I_x = \frac{V - V_T}{3} \Rightarrow V - V_T = 3I_x \quad (2)$$

$$I_x + 1 = \frac{V_T}{4} \Rightarrow 4I_x + 4 = V_T \quad (3)$$

$$(1) \Rightarrow (2) \Rightarrow 2.5I_x - V_T = 3I_x \Rightarrow V_T = -0.5I_x \quad (4)$$

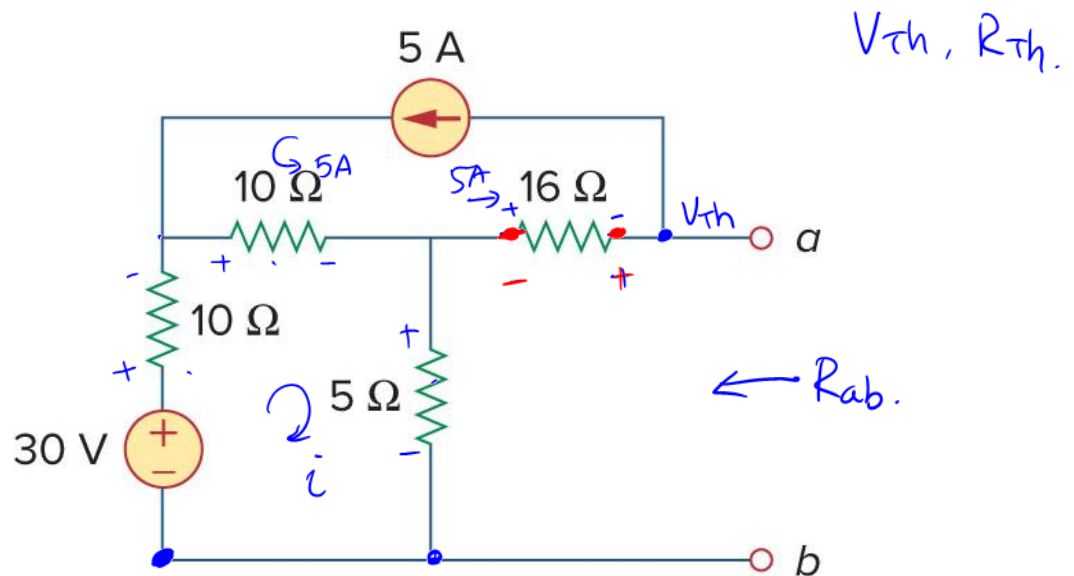
$$(4) \Rightarrow 8V_T = -4I_x \Rightarrow 4I_x = -8V_T. \quad (5)$$

$$(5) \Rightarrow (3): -8V_T + 4 = V_T \Rightarrow 9V_T = 4$$

$$\Rightarrow V_T = \frac{4}{9} V. \quad R_{Th} = \frac{V_T}{1A} = \frac{4}{9} \Omega.$$

Example 10: Thévenin

- Obtain the Thévenin equivalent circuit with respect to terminals a - b .



$$R_{th} = 16 + 5 \parallel 20 = 16 + 4 = 20 \Omega.$$

$$\text{KVL: } -30 + 10i + 10(5+i) + 5i = 0$$

$$25i = 30 - 50 = -20 \Rightarrow i = -0.8 \text{ A.}$$

$$V_{th} = -5 \times 16 + 5i = -84 \text{ V.}$$